

Multiple Choice

1. **Answer:** A

Options: A) 4; B) 1; C) 256; D) 16

Note: Correct option: A - 4

2. **Answer:** D

Options: A) All bit strings whose number of zeros is a multiple of five; B) All bit strings starting with a zero and ending with a one; C) All bit strings with an even number of zeros; D) All bit strings with more ones than zeros

Note: Correct option: D - All bit strings with more ones than zeros

3. **Answer:** B

Options: A) /; B) //; C) %; D) |

Note: Correct option: B - //

4. **Answer:** D

Options: A) 1; B) 2; C) 3; D) 4

Note: Correct option: D - 4

5. **Answer:** D

Options: A) 53; B) 29; C) 50; D) 100

Note: Correct option: D - 100

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1/K'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'O(1)'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'one in which each main memory word can be stored at any of 3 cache locations'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'O(log log N)'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def search_literal(pattern,text): | return (s, e)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def find_Digits(n): | return 0; | return 1; | return math.floor(x) + 1;`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key